Week 12, Lecture 22
Competitive Equilibrium with Arrow Securities

Assigned reading: QuantEcon, *Competitive Equilibria with Arrow Securities*, https://python.quantecon.org/ge_arrow.html.

1 From two periods to an infinite horizon

We extend Lecture 1 to infinitely many periods and a finite Markov state space $S = \{1, \dots, n\}$. The state s_t evolves with transition matrix P:

$$P_{ij} = \Pr(s_{t+1} = j \mid s_t = i).$$

There are K consumers, indexed by k, with state-dependent endowments $y^k(i)$. All have the same time discount factor $0 < \beta < 1$, beliefs P, and CRRA period utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t^k), \qquad u'(c) = c^{-\gamma}, \quad \gamma > 0.$$

Here $u(c) = c^{1-\gamma}/(1-\gamma)$ when $\gamma \neq 1$, and $u(c) = \ln c$ when $\gamma = 1$. Aggregate endowment in state i is

$$\boxed{y(i) = \sum_{k=1}^{K} y^k(i).}$$

Consumers differ in their endowment exposures but can trade contingent claims to share risk.

2 Sequential Arrow securities

When the current state is i, a one-period Arrow security for next state j costs Q_{ij} units of today's consumption and pays one unit of consumption next period if $s_{t+1} = j$ (zero otherwise). Let Q be the $n \times n$ matrix of such prices.

A risk-free claim paying one unit in every next-period state is a portfolio of one unit of each Arrow claim. Its price is $\sum_j Q_{ij}$, so its gross risk-free return is

$$\boxed{R_i = \frac{1}{\sum_j Q_{ij}}.} \tag{1}$$

3 The household's recursive problem

Let a denote the household's current financial payoff, and $\hat{a}(j)$ its purchased payoff for next period if state j occurs. The state-i budget constraint is

$$\boxed{c + \sum_j Q_{ij}\hat{a}(j) = y^k(i) + a.} \tag{2}$$

The household's Bellman equation is

$$v^k(a,i) = \max_{c,\{\hat{a}(j)\}} \left\{ u(c) + \beta \sum_j P_{ij} v^k(\hat{a}(j), j) \right\},$$

subject to (2) and admissible debt limits. The natural debt limits prevent consumers from borrowing more than the market value of their future endowment stream.

4 The Arrow-pricing equation

The first-order condition for $\hat{a}(j)$ and the envelope condition for financial wealth yield

$$\boxed{Q_{ij} = \beta P_{ij} \frac{u'(c^k(j))}{u'(c^k(i))}.} \tag{3}$$

An Arrow claim is valuable when it pays in a state with high marginal utility. For CRRA preferences,

$$\boxed{Q_{ij} = \beta P_{ij} \left(\frac{c^k(j)}{c^k(i)} \right)^{-\gamma}.} \tag{4}$$

The expression assumes stationary Markov equilibrium consumption policies, and the comparison is relevant for transitions with $P_{ij} > 0$.

5 Complete markets and constant consumption shares

All consumers face the same Q_{ij} and P_{ij}. Thus, for any two consumers k, m and any transition with positive probability, (4) implies

$$\left(\frac{c^k(j)}{c^k(i)} \right)^{-\gamma} = \left(\frac{c^m(j)}{c^m(i)} \right)^{-\gamma}.$$

Consequently,

$$\frac{c^k(j)}{c^m(j)} = \frac{c^k(i)}{c^m(i)}.$$

In a connected state space, relative consumption is constant across states. Goods-market clearing, $\sum_k c^k(i) = y(i)$, then gives

$$\boxed{c^k(i) = \alpha_k y(i), \qquad \alpha_k > 0, \quad \sum_k \alpha_k = 1.} \tag{5}$$

Complete markets insure individual endowment fluctuations; aggregate consumption fluctuations remain and are shared proportionally.

6 Pricing from aggregate endowment

Substituting (5) into (4), the consumption share cancels:

$$\boxed{Q_{ij} = \beta P_{ij} \left(\frac{y(j)}{y(i)} \right)^{-\gamma}.} \tag{6}$$

Key insight: The equilibrium Arrow-price matrix is determined by aggregate endowment, its Markov transition probabilities, discounting, and risk aversion. We can compute Q before determining individual consumption shares.

7 Valuation, natural debt limits, and financial wealth

Define the valuation matrix

$$\boxed{V = (I - Q)^{-1} = I + Q + Q^2 + \cdots,} \tag{7}$$

assuming the series converges (equivalently, the spectral radius of Q is less than one). The matrix V here is a *valuation operator*, not an indirect utility function.

For any state-dependent income vector z, the vector Vz gives its present value, including its current-period payoff. In particular, the present value of consumer k's endowment stream is

$$\boxed{\bar{A}^k = Vy^k.}$$

The negative of this endowment value gives the state-dependent natural debt limit under the usual nonnegative-consumption interpretation.

The financial wealth needed to support consumer k's insured consumption stream is

$$\boxed{\psi^k = V(\alpha_k y - y^k).} \tag{8}$$

Indeed, $\psi^k = \alpha_k y - y^k + Q\psi^k$, or equivalently,

$$\alpha_k y(i) + \sum_j Q_{ij} \psi^k(j) = y^k(i) + \psi^k(i),$$

which is exactly the household budget equation when current financial wealth is $\psi^k(i)$ and the household purchases next-period claims $\psi^k(j)$.

8 Determining consumption shares

Suppose all consumers begin with zero financial wealth and the initial state is s_0. Then $\psi^k(s_0) = 0$. From (8),

$$[V(\alpha_k y - y^k)]_{s_0} = 0,$$

so

$$\boxed{\alpha_k = \frac{[Vy^k]_{s_0}}{[Vy]_{s_0}}.} \tag{9}$$

Thus each consumer's consumption share equals the fraction of aggregate endowment wealth owned initially. Summing (9) over consumers gives one.

The equilibrium computation follows the sequence

$$\boxed{(P, y^1, \ldots, y^K, \beta, \gamma) \longrightarrow Q \longrightarrow V \longrightarrow \alpha_k \longrightarrow (c^k, \psi^k).}$$

9 A two-state numerical illustration

Consider two states and transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}, \qquad \beta = 0.98.$$

Suppose aggregate endowment is constant, $y(1) = y(2) = 1$, even though the consumers' individual endowments vary inversely across states. Then $y(j)/y(i) = 1$, so (6) becomes

$$\boxed{Q = \beta P = \begin{pmatrix} 0.49 & 0.49 \\ 0.49 & 0.49 \end{pmatrix}.}$$

The price of a one-period risk-free claim is $0.49 + 0.49 = 0.98$, and

$$\boxed{R = \frac{1}{0.98} \approx 1.0204.}$$

Each consumer consumes a constant share α_k of aggregate endowment, so $c^k(1) = c^k(2) = \alpha_k$. Individual consumption is fully insured against state variation. The exact shares depend on the values of the consumers' endowment streams and the initial state via (9).

10 Sequential markets and Arrow–Debreu markets

In an Arrow–Debreu arrangement, claims to consumption contingent on entire future histories are traded at date 0. In sequential Arrow markets, only one-period contingent claims trade at each date, and portfolios are rebalanced as new information arrives. With complete markets, appropriate solvency conditions, and correct expectations of future prices, both arrangements support the same allocations. Recursive methods let us study the sequential arrangement using state-dependent policies and prices.

11 Conclusions

1. Complete Arrow markets allow agents to insure idiosyncratic endowment risk.
2. Identical CRRA preferences imply constant shares of aggregate consumption.
3. Arrow prices reflect probabilities, discounting, risk aversion, and aggregate consumption growth.
4. The valuation operator $(I - Q)^{-1}$ prices endowment streams and supports the calculation of natural debt limits, financial wealth, and consumption shares.
5. Sequential trading of Arrow securities implements the complete-markets allocation.